9/29/26

1. Team Info

A. Lauren Kokesh: Project Manager

B. Kaeden Kiffe: Developer

C. Jacob McDonald: Developer

D. Isaac Montanes: Developer

Links to project documents:

lkokesh23/Bruin-Scheduling-System: CSIS 321 Software Engineering Project

https://trello.com/b/zn5QOvHT/software-engineering

Communication Tools: Email and Text

2. Project description

The goal of our project is to create a standardized scheduling system that formats the user's schedule into a Google Doc based on user input and focuses on readability. It will be designed with a simple GUI and user-focused features with the goal of not overcomplicating a simple task. We will utilize the Google API for Java to achieve Google product compatibility. Whether it be a professor planning his class and office schedule, or a student trying to manage their busy day, schedules are a vital part of George Fox University. Without a university standard scheduling system, it is harder to plan and collaborate with others, not knowing what times and days they have availability. Through discussion with university faculty and other members, we aim to fix the inconsistent scheduling systems through a new and easy-to-use interface.

Currently, our stakeholder, a Theology Teacher's Assistant, formats everything manually by hand and cites it as more time-consuming than the rest of his tasks. Current practice is limited by having to double-check and maintain consistency, and the process of dealing with multiple

Google systems. Our approach will feature a simple GUI interface, where the user sets up the basic information for their Doc, inputs their scheduling preferences, and tweaks the styling preferences. The program will set up the Doc, save it, and forward it automatically to anyone the user wants it shared with.

When our program is complete and we hand it to our stakeholder, as well as any Fox student, staff, or TA interested, we will have created a tool capable of setting a standard scheduling system. This will make communication easier, especially across field divides such as STEM and the Arts. As part of our approach, we plan to use the Google API for Java to integrate Google Docs and Google Drive usage. As part of the program, we will also try to implement an ability to save settings and preferences for future sessions. We will also work with the community to find ways to make it easier for every user to use our product, and make sure to be as color-blind accommodating as possible in GUI design.

We face several challenges with developing this software. The first one is choosing the scope of the GUI. Having one that is too simple or is underdeveloped takes away from functionality and user experience. Spending excess time designing everything would involve unnecessary complexity and would be an exercise in futility. The second challenge hinges on how cooperative the Google API is to integrate, test, and understand. The final risk revolves around proper organization; as long as we can set weekly deadlines for what we need done as a structure to plan around, we can accurately assess focus for each person for that week or day.

Major Features:

- GUI Interface
- Google Docs Integration & Google API
- Use of a Calendar and a Clock

- Storage Database

Stretch Goals:

- George Fox Login Page
- Community-wide Access

3. Use Cases (Functional Requirements)

Use Case 1:

A. Actors: Mr. Ancheta (TA & Stakeholder)

B. Triggers: The need to organize a meeting between a professor and a student.

C. Preconditions: Having a GFU email

D. Postconditions: Meeting is scheduled between the professor and student, and both are notified about it.

E. List of Steps:

   a. Log in to the system with a GFU Email

   b. Navigate to “schedule meeting”

   c. Select the participants of the meeting

   d. System sends request to add meeting to their calendars

   e. Meeting is accepted and is now visible on their calendars

F. Extensions/Variations of the success scenario

   a. System flags a schedule conflict, Actor must select a different time

G. Exceptions: Failure conditions and scenarios

   a. The meeting is rejected

   b. The student/professor leaves a note on why it was rejected

   c. They work together to find a time that works for both of them.

Use Case 2:

A. Actors: Any GFU student/professor/faculty member
B. Triggers: The desire to create a personal weekly schedule
C. Preconditions: Having a GFU email
D. Postconditions: A clear, simple weekly schedule is available in Google Docs
E. List of Steps:
    a. Log in to the system with a GFU email
    b. Navigate to “Create weekly schedule”
    c. Fill out the forms with class times, meetings, and other personal items
    d. System previews the formatted schedule
    e. Actor selects things to change if needed
    f. Select people to share the schedule with
    g. Schedule is sent to their Google Drive and shared with select people
F. Extensions/Variations of the success scenario:
    a. Actor imports a previous schedule to make edits to
    b. Actor saves the list of people to share it with, to send a schedule every week
G. Exceptions: Failure conditions and scenarios:
    a. Actor enters conflicting times; the system flags it and requests clarification
    b. The system cannot share with all the people due to privacy selections and other restrictions.

Use Case 3:

A. Actors: Any GFU student/professor/faculty member
B. Triggers: The need to decline an event/meeting invitation

C. Preconditions: Having a GFU email

D. Postconditions: Event created in actor's schedule

E. List of Steps:

    a. Actor receives event/meeting invitation

    b. Actor accepts meeting

    c. Event is added to their calendar and/or pre-existing schedule

F. Extensions/Variations of the success scenario:

    a. Actor requests a reschedule of the meeting

    b. Event is rescheduled to a different date and time

G. Exceptions: Failure conditions and scenarios:

    a. Failure to remember the event date and time

    b. Miscommunication of when the rescheduled event is

    c. User has an emergency and is unable to make the event

Use Case 4:

A. Actors: Any Professor at GFU

B. Triggers: The need to have a meeting with a student regarding grades.

C. Preconditions: The student is at GFU and has a GFU email.

D. Postconditions: A meeting time is scheduled and followed through on.

E. List of Steps:

    a. Actor realizes a student is not passing exams and decides to set up a meeting.

    b. Actor schedules a meeting and sends a notification to the student regarding times to meet.

    c. Student accepts the meeting time.

F. Extensions/Variations of the success scenario:

    a. The time doesn't work, so the student picks a different time with the actor.

G. Exceptions: Failure conditions and scenarios:

    a. Student fails to receive notification.

    b. A time doesn't work between the actor and the student.

4. Non-functional Requirements

Describe at least three non-functional requirements, e.g., related to scalability, usability, security, and privacy, etc.

1. Security and Privacy: Using a George Fox email
2. Scalability: It should be able to handle use by a large portion of the university.
3. Usability: Simple and clear interface

5. Team process description

Describe your quarter-long development process.

Specify and justify the software toolset you will use:

- We will be using Java and integrating it with the Google API. This will ensure accessibility because everyone has access to Google Docs.

Define and justify each team member's role: why does your team need this role filled, and why is a specific team member suited for this role?

- Project Manager: Lauren Kokesh - we need this role to have someone making sure everything is getting done on time and to organize the developers' efforts.
- Developer(s): Kaeden Kiffe, Jacob McDonald, Isaac Montanes - we need people who are focused on creating features of the product and can spend time thinking on how to best implement the feature.

Provide a schedule for each member (or sub-group) with a measurable, concrete milestone for each week. “Feature 90% done” is not measurable, but “use case 1 works, without any error handling” is.

- Kaeden: Start the design process for the Scheduling System.
- Lauren: Work on defining set deadlines for each team member.
- Jacob: Start the design process for the Scheduling System.
- Isaac: Research the pros and cons between an app vs. a website.

Specify and explain at least three major risks that could prevent you from completing your project.

- Risk 1: Missing deadlines and having incomplete assignments.
- Risk 2: Trying to incorporate too many aspects into our project.
- Risk 3: We take on a task that is too difficult and requires a more complex skill set than we currently possess.

Describe at what point in your process external feedback (i.e., feedback from outside your project group, including the project manager) will be most useful and how you will get that feedback.

- The point in the process when external feedback will be most useful to us is in the first couple of weeks, when we are still building the shell of our project and can easily update and change design specifications. We will get this feedback by going to office hours and discussing the project with peers and GFU students.